WashU McKelvey Engineering

Spring 2026 MEMS 3430 Design of Thermal Systems

Design Homework #2 - 2nd Submission

Rankine Bottoming Cycle

Assumptions and Boundary Conditions

Assumptions.

(1) The temperature of the Mississippi is assumed to remain at a constant value, a simplification of the real world necessary for a snapshot calculation of thermal properties.
(2) Minor losses in the condenser tubes and losses from the pumps supplying the coolant liquid are neglected, as well as any friction losses of the steam in the condenser.
(3) It is assumed that the condenser is an ideal component, with constant entropy and an isentropic efficiency of 100%. This would not be the case in a real condenser, but this simplifies the problem for ease of calculation.
(4) The Rankine and coolant pumps are assumed to be perfectly efficient and there are no additional losses caused by the pumps. This is a valid simplification of the problem, while a real world system would have additional losses due to pump efficiency.
(5) It is assumed that there are no losses in the heat exchanger between the topping and bottoming cycle, which is a valid assumption for a perfectly insulated heat exchanger but a real system would experience some heat losses.
(6) It is assumed that there are no pressure losses through the condenser and the heat exchanger, such that the pressure is constant between those states. This is a valid assumption for a ideal Rankine cycle.
(7) It is assumed that the system is operating in steady-state conditions, and does not include any startup or shutdown behavior that would reduce the performance of the cycle. This is a valid simplification of the problem such that the design can accurately model the majority of the behavior of the system.
(8) It is assumed that the fluid leaving the condenser is a saturated liquid, which is a valid assumption for an ideal condenser that transforms the entirety of the steam generated through the turbine to condensed liquid.

Boundary Conditions.

(1) The temperature rise of the coolant water across the condenser is set to be 21°F, or 11.67 K [1].
(2) The temperature of the Mississippi in Alton, Illinois was used for the snapshot temperature of the coolant water, set to be 12°C, or 285.15 K [2]. Using the maximum temperature rise of the coolant water of 11.67 K, the temperature of the coolant water when it exits the condenser is therefore 296.82 K.
(3) The fluid in the condenser from the Rankine cycle must be at a higher temperature than the coolant water in order for heat transfer to occur. Therefore, the same temperature rise will be used as a pinch point for the fluid in the condenser, such that T_9 = 308.49 K.
(4) The inlet temperature to the steam turbine T_7 must be lower than the outlet temperature of the gas turbine in the topping cycle to allow for heat transfer to occur in the right direction. A literature review was performed in order to determine an acceptable value for a common pinch point temperature, which was found to be 8°C. From the first design homework, T_4 was found to be 260.7°C, such that T_7 = 260°C − 8°C = 525.85°C [3].

(5) The turbine was given to have an isentropic efficiency of 90%. A literature search to find an acceptable minimum quality was performed, resulting in a minimum turbine exit quality of 92% in order to prevent erosion from droplet impingement on the turbine blades [4].
(6) The condenser is a shell & tube heat exchanger with 1000 parallel tubes, each with a 25mm internal diameter and a 10m length. Friction losses of the coolant water from flow through the condenser heat exchanger pipes was calculated, using a relative roughness of 0.045mm [5].
(7) The high temperature heat exchanger provides 30 MW of thermal energy from the Brayton topping cycle, which is passed into the Rankine bottoming cycle through the same heat exchanger.

Discussion of environmental effects of power plant thermal discharge of the cooling water. The reason why the boundary condition that limits the output of the coolant water back into the system can be made is because of the distinct environmental impact that a extremely heated coolant would have back on the natural system it was taken from. This effect is known as thermal pollution, where human activities change temperature of the water, negatively affecting fish and aquatic life and altering the waters chemistry. Common impacts of thermal pollution included reduced reproductive success of fish and aquatic life, habitat degradation, and fish kills in extreme cases [6]. Warmer water also holds less oxygen than warmer water, so infusion of heat in a body of water reduces its levels of available oxygen for oxygen-breathing creatures [7]. Algae, especially blue-green algae, thrive in warmer water, causing algae blooms that absorb oxygen and sunlight, creating huge areas where aquatic creatures and plant life can no longer survive due to the degraded conditions [7]. It is therefore absolutely necessary to follow the guidelines set by experts in order to product the natural habitats needed by our ecosystems and to prevent these kinds of environmental disasters from occurring.

Schematics and Diagrams

Figure 1 shows a schematic of the combined cycle gas turbine (CCGT) which includes the Rankine bottoming cycle being modeled in this assignment. In the bottoming cycle, 30 MW of heat from the Brayton topping cycle enters through the heat exchanger between states 6 and 7. The fluid in state 7 moves through a steam turbine and is expanded to state 8, producing turbine work in the process.

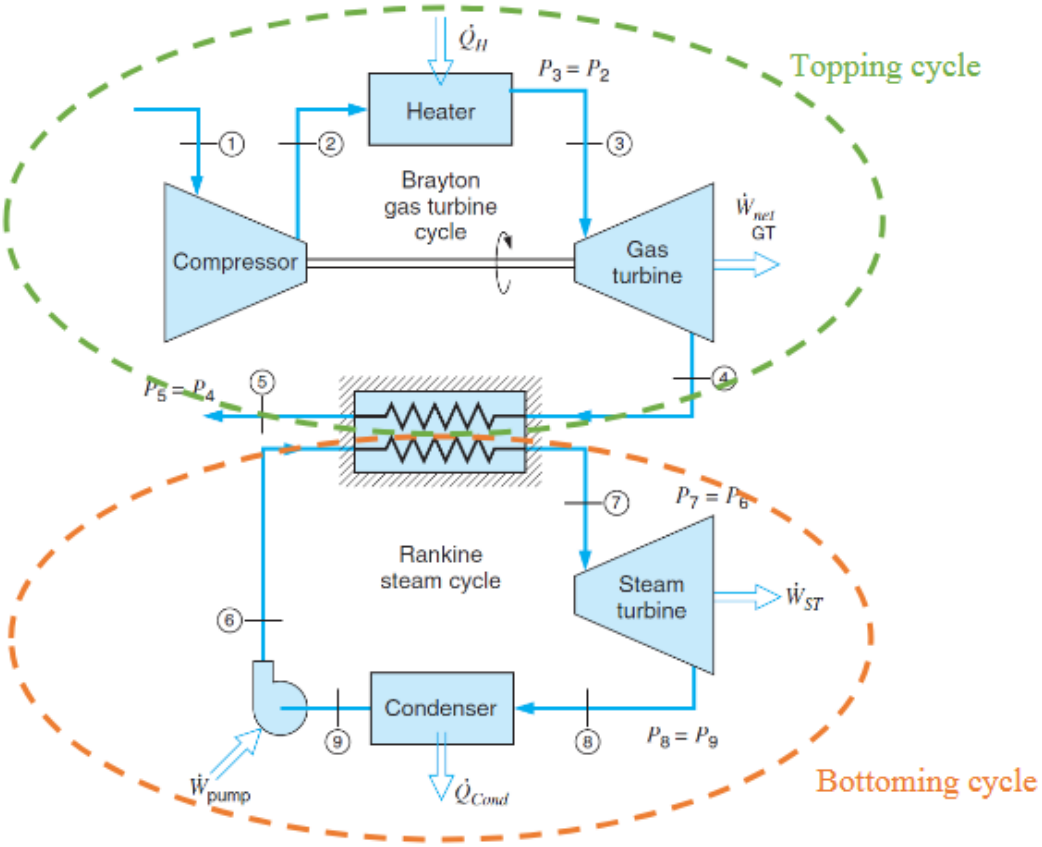


Figure 1 Schematic of the Combined Cycle Gas Turbine

The fluid at state 8 then moves into a condenser, where the steam condenses on the outside of a shell & tube heat exchanger, inside of which coolant water is fed through a heat exchanger that carries heat out of the system. Liquid water at state 9 at the exit of the condenser is then fed into a pump, which uses work to compress the liquid to state 6. The cycle then repeats as a closed system. Figure 2 shows the system inside the condenser supplying coolant water.

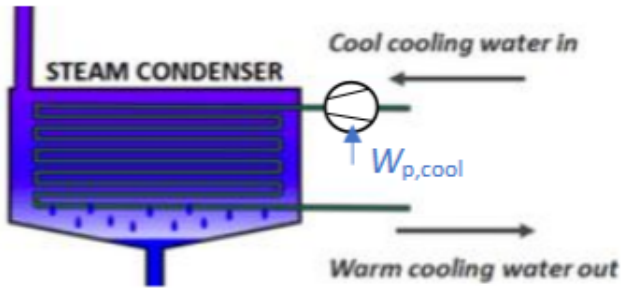


Figure 2 Shell & Tube Heat Exchanger System Inside Condenser

Steam condenses on the outer surface of the 1000 parallel tubes inside the condenser, which each carry coolant water through the system. This coolant is supplied via a pump, which requires power to operate. Figure 3 shows a temperature - entropy diagram of the Rankine bottoming cycle.

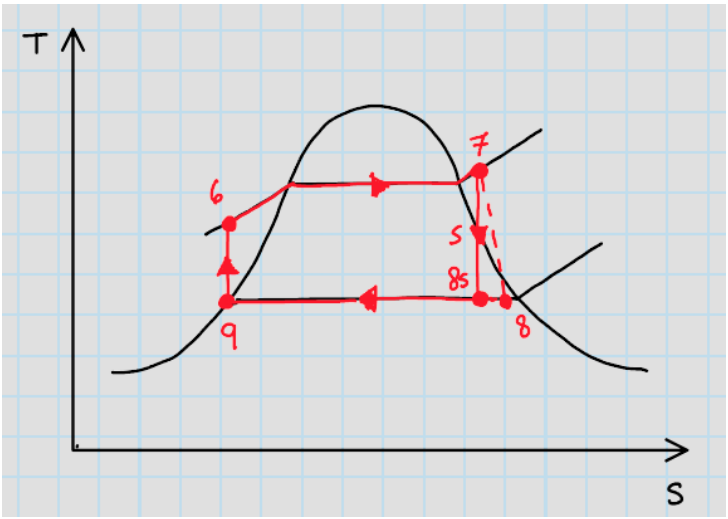


Figure 3 T-s Diagram of the Rankine Bottoming Cycle

In the cycle, states 6 to 7 experience isobaric heating, states 7 to 8 experience adiabatic expansion, states 8 to 9 experience isobaric cooling, and states 9 to 6 experience adiabatic compression. The states representative of the coolant water traveling through the condenser are neglected for the purposes of the overall T-s diagram of the Rankine bottoming cycle.

Equations and Calculations

The pressure and enthalpy after the condenser can be determined from the set temperature of 308.49 K, and the fact that the state after the condenser can be assumed to be a saturated liquid. The temperature of the inlet of the condenser is set to be 285.15 K, the temperature of the Mississippi River on 3/2/2026 at the time of checking. Using the pinch point researched, the outlet temperature is then 296.82 K. Using these temperatures, the enthalpies of the inlet and outlet states can be determined, as well as the density and dynamic viscosity of water at the average temperature of the coolant. Equation 1 is used to find the enthalpy at state 8:

$$h_8 = h_7 - \eta_t(h_7 - h_{8s}) \tag{1}$$

Where h_8 is the enthalpy at state 8 in kJ/kg, η_t is the isentropic efficiency of the turbine, 0.90, h_7 is the enthalpy at state 7, and h_{8s} is the isentropic enthalpy at state 8. The information for state 7 is determined using the pressure and temperature set from the boundary conditions.
In order to determine the enthalpy of state 6, a pump equation was utilized, shown below in Eq. 2

$$h_6 = v_9(P_7 - P_9) + h_9 \tag{2}$$

Where v_9 is the specific volume of the fluid at state 9, calculated from Eq. 3 in m^3/kg, P_7 is the pressure at state 7, and P_9 is the pressure at state 9, both in Pa.

$$v_9 = \frac{1}{\rho_9} \tag{3}$$

Once h_6 is determined, the mass flow rate of the bottoming cycle in kg/s can be found using Eq. 4:

$$\dot{m}_{rankine} = \frac{\dot{Q}_H}{h_7 - h_6} \tag{4}$$

Where $\dot{Q}_H$ is the 30 MW from the topping cycle. Once this mass flow rate is calculated, the next step is to determine the heat needed to be dissipated by the condenser, shown in Eq. 5

$$\dot{Q}_c = \dot{m}_{rankine}(h_8 - h_9) \tag{5}$$

Where $\dot{Q}_c$ is in kW, and the enthalpies are in kJ/kg. Using an energy balance, it can be shown that the same heat that is dissipated by the steam condensing is taken out by the coolant water leaving the system, which can then be used to solve for the mass flow rate necessary to carry that much hate out of the system. This is shown in Eq. 6

$$\dot{m}_{coolant} = \frac{\dot{Q}_c}{h_o - h_i} \tag{6}$$

Where h_o and h_i are the enthalpies of the outlet and inlet states of the coolant cycle in kJ/kg, respectively. This can be split into the mass flow rate per pipe using Eq. 7:

$$\dot{m}_{coolant,pipe} = \frac{\dot{m}_{coolant}}{N} \tag{7}$$

Where N represents the total number of parallel tubes in the condenser, which is 1000. In order to find the velocity in each of the tubes, the following relationship shown in Eq. 8 can be written:

$$V = \frac{\dot{m}_{coolant,pipe}}{\rho_{coolant} A} \tag{8}$$

Where V is in m/s, and A represents the area of one of the tubes, which is $\pi\frac{D^2}{4}$. This velocity can then by used to find the Reynolds Number in Eq. 9

$$Re = \frac{\rho_{coolant} V D}{\mu_{coolant}} \tag{9}$$

If this Reynold's number is below 2300, then Eq. 10 can be used to find the friction factor f for laminar flow:

$$f = \frac{64}{Re} \tag{10}$$

If the Reynold's number is above 2300, which neglects the transition phase, then Eq. 11 can be used to find the friction factor for turbulent flow:

$$\frac{1}{\sqrt{f}} = -2\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right) \tag{11}$$

Here, ϵ is the pipe roughness, found in the boundary conditions for commercial steel of 0.045mm. This friction value can then be used to find the change in pressure ΔP in Pa that occurs in the pipes, shown in Eq. 12:

$$\Delta P = f\frac{L}{D}\rho\frac{V^2}{2} \tag{12}$$

Where L is the length of a pipe in meters, D is the diameter of a pipe in meters, ρ is the density of the water at the average temperature in kg/m^3, and V is the velocity of the water in m/s. This pressure change can be used to find the power required to run the coolant pipe in Eq. 13:

$$\dot{W}_{pipe,coolant} = \frac{\dot{m}_{coolant}\Delta P}{\rho_{coolant}} \tag{13}$$

Which is in units of kW. Going back to the Rankine cycle pump, which is necessary to find the net work, it can be found using an energy balance on the pipe, shown in Eq. 14

$$\dot{W}_{pump,rankine} = \dot{m}_{rankine}(h_6 - h_9) \tag{14}$$

Which is in units of kW. Similarly, the power generated by the turbine can be found in Eq. 15:

$$\dot{W}_t = \dot{m}_{rankine}(h_7 - h_8) \tag{15}$$

The turbine generated by the turbine minus the total power needed to run the pumps in the cycle results in the net power of the cycle, shown in Eq. 16

$$\dot{W}_{net} = \dot{W}_t - \dot{W}_{pump,coolant} - \dot{W}_{pump,rankine} \tag{16}$$

Which is in units of kW. This can then be used to find the thermal efficiency of the Rankine cycle in Eq. 17

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_H} \tag{17}$$

Finally, the total thermal efficiency of the CCGT can be found by adding the total net output of the cycle minus the total heat leaving the cycle, shown in Eq. 18

$$\eta_{th,CCGT} = \frac{\dot{W}_{gas,net} + \dot{W}_{steam} - \dot{W}_{pumps}}{\dot{Q}_H} \tag{18}$$

Results and Discussion

The goal was to optimize the thermal efficiency of the cycle by varying the turbine inlet pressure, T_7. This was accomplished by iterating through all pressure values between a minimum of 1×10^5 Pa and the saturation pressure at state 7, given that it has a temperature of 525.85 K. This boundary condition is possible because of the restriction that the turbine inlet cannot be in the two-phase. Figure 4 shows the plot of thermal efficiency versus turbine inlet pressures, with an additional y-axis showing the quality of the fluid after moving through the turbine.

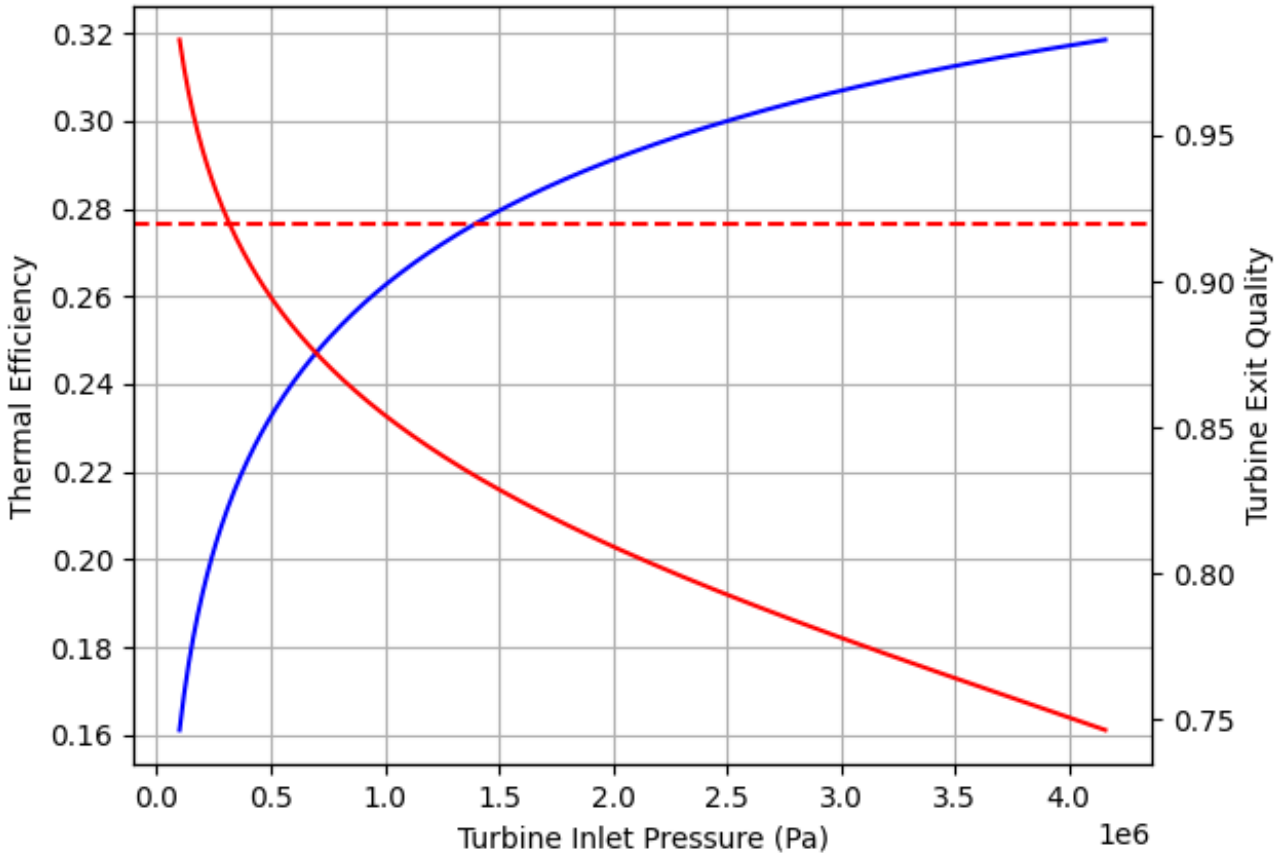


Figure 4 Thermal Efficiency versus Turbine Inlet Pressure and Turbine Outlet Quality

The dashed line in red represents the minimum exit quality of 92%, the solid line in red represents the quality of the exit, and the solid line in blue represents the thermal efficiency of the cycle, all with respect to the turbine inlet pressure. From the plot, it can be seen that the highest thermal efficiency occurs at the farthest right point before the quality goes below 0.92. The maximum possible thermal efficiency shown is just below 0.32, but is at a quality of just above 0.75, much lower than the minimum recommended via a literature search. While this might be more efficient, in the long run the significant damage this would pose to the turbine blades would make it more costly to maintain. The highest quality occurs just under 100%, but at this quality the thermal efficiency of the cycle is around 0.16, meaning that there is so much waste from the cycle that it is generating almost no useful work via the turbine. At the optimal point, the following final answers were determined:

(1) Rankine cycle mass flow rate: $\boxed{10.62 \text{ kg/s}}$
(2) Coolant loop mass flow rate: $\boxed{484.22 \text{ kg/s}}$
(3) Final turbine inlet pressure: $\boxed{0.31 \text{ MPa}}$
(4) Turbine power output: $\boxed{6352.39 \text{ kW}}$
(5) Rankine pump input: $\boxed{3.25 \text{ kW}}$

(6) Coolant pump input: $\boxed{2.69 \text{ kW}}$
(7) Final total net power output: $\boxed{6346.45 \text{ kW}}$
(8) Final thermodynamic efficiency of the cycle: $\boxed{0.2116}$
(9) Total thermodynamic efficiency of the entire CCGT: $\boxed{0.3714}$

Sanity Check. Comparing the results of the first and second DHW, the Brayton topping cycle was able to generate 112.32 MW of power through its solar plant, while the Rankine steam cycle is only able to generate 6.346 MW of power. Realistically, this makes sense, because the goal of the CCGT is to use the heat generated by the Brayton cycle to evaporate steam for the Rankine cycle, therefore increasing overall efficiency. Based on the lecture notes , I expected the steam cycle to have a higher efficiency than the gas cycle, but in my first DHW my thermal efficiency was found to be 0.3515, over 10% higher than the efficiency of the steam cycle. However, the final efficiency was found to be 0.37135, indicating that the final efficiency of the entire CCGT is higher after using the preheated steam from the topping cycle. This is an indication that the values calculated were physically reasonable, considering that the overall final answers followed the correct patterns. The coolant loop mass flow is very large compared to the overall Rankine cycle, but the reason is that the enthalpies used for the coolant water are so low that in order to dissipate all of the heat generated by the cycle, the mass flow rate has to be extremely large. This is therefore not a concerning value. The overall pressure in the turbine is on the lower end of the plot, however the selected turbine exit quality value is high at 92%, limiting most of the plot, and the overall cycle is not extremely efficient because of the decision during the first DHW to maximize the power output of the cycle rather than maximize the thermal efficiency.

References

[1] n.p., 1992, “Public Notice and Fact Sheet/Rationale, Thermal Discharges,” T.H. Allen Fossil Plant.

[2] n.p., n.d., “Water Temperature of the Mississippi River in Alton, IL,” watertemps.com, accessed 3/3/2026, https://watertemps.com/Alton/IL/US/ALNI2

[3] n.p., n.d., “Heat exchanger appraoch temperature,” EnggCyclopedia, accessed 3/3/2026, https://enggcyclopedia.com/2019/05/heat-exchanger-approach-temperature/

[4] Muller-Steinhagen, H. M. G., 2011, “Rankine Cycle,” Thermopedia, accessed 3/3/2026, https://www.thermopedia.com/content/1072/

[5] n.p., 2003, “Surface Roughness Coefficients,” The Engineering Toolbox, accessed 3/4/2026, https://www.engineeringtoolbox.com/surface-roughness-ventilation-ducts-d_209.html

[6] n.p., n.d., “Thermal Pollution,” Dane County Land Water Resources Department, accessed 3/3/2026, https://lwrd.danecounty.gov/what-we-do/Pollution-and-Solutions/Thermal-Pollution

[7] n.p., 2024, “What Is Thermal Pollution? Causes, Impact Solutions,” perch energy, accessed 3/3/2026, https://www.perchenergy.com/blog/environment/what-is-thermal-pollution-causes-impact

Appendix: Python Code

```
# Aidan Moriarty
# DHW-2 Optimizer Program
# 527029

import CoolProp.CoolProp as CP
import math
import numpy as np
import matplotlib.pyplot as plt

# Params
Q_H_dot = 30e3 # kW
N = 1000
D = 0.025
A = (math.pi * D**2) / 4
L = 10
epsilon = 0.045  # mm
min_x = 0.92

T_7 = 525.85
eta_t = 0.90

P_min = 1e5
P_max = CP.PropsSI('P', 'T', T_7, 'Q', 1, 'Water')
# P_max = 1e7
dP = 10000

P_vals = np.arange(P_min, P_max, dP)
eta_vals = []
x_vals = []
m_dot_vals = []
m_dot_coolant_vals = []
t_power_vals = []
rankine_pump_vals = []
coolant_pump_vals = []
net_power_vals = []

T_9 = 308.49 # K
P_9 = CP.PropsSI('P', 'T', T_9, 'Q', 0, 'Water') # Pa
h_9 = CP.PropsSI('H', 'T', T_9, 'Q', 0, 'Water') # J / kg

T_i = 285.15
ΔT = 21 * 5/9
T_o = T_i + ΔT

h_i = CP.PropsSI('H', 'T', T_i, 'Q', 0, 'Water') # J / kg
h_o = CP.PropsSI('H', 'T', T_o, 'Q', 0, 'Water') # J / kg

T_avg = (285.15 + 295.37)/2 # K

rho = CP.PropsSI('D', 'T', T_avg, 'Q', 0, 'Water') # kg / m^3
mu = CP.PropsSI('V', 'T', T_avg, 'Q', 0, 'Water')

for P_7 in P_vals:

```

```
    s_7 = CP.PropsSI('S', 'P', P_7, 'T', T_7, 'Water') # J/kg*k
    h_7 = CP.PropsSI('H', 'P', P_7, 'T', T_7, 'Water') # # J / kg

    h_8s = CP.PropsSI('H', 'P', P_9, 'S', s_7, 'Water') # J / kg
    h_8 = h_7 - eta_t * (h_7 - h_8s) # J / kg

    x_8 = CP.PropsSI('Q', 'P', P_9, 'H', h_8, 'Water')
    x_vals.append(x_8)

    rho_9 = CP.PropsSI('D', 'P', P_9, 'H', h_9, 'Water')
    v_9 = 1 / rho_9
    h_6 = v_9 * (P_7 - P_9) + h_9 # J / kg

    m_dot = Q_H_dot / ((h_7 - h_6)/1000) # kg / s
    Q_c_dot = m_dot * ((h_8 - h_9)/1000) # kW

    m_dot_coolant_total = Q_c_dot / ((h_o - h_i)/1000) # kg / s
    m_dot_coolant_pipe = m_dot_coolant_total / N

    m_dot_vals.append(m_dot)
    m_dot_coolant_vals.append(m_dot_coolant_total)

    V = m_dot_coolant_pipe / (rho * A)
    Re = (rho * V * D) / mu

    if Re < 2300:
        f = 64 / Re
    else:
        f = (-1.8 * math.log10((epsilon/(D*1000)/3.7)**1.11 + 6.9/Re))**-2

    P_pipe = f * (L/D) * rho * (V**2/2)

    W_dot_pump_coolant = m_dot_coolant_total * (P_pipe / rho) / 1000
    W_dot_pump_rankine = m_dot * ((h_6 - h_9)/1000) # kW

    coolant_pump_vals.append(W_dot_pump_coolant)
    rankine_pump_vals.append(W_dot_pump_rankine)

    W_dot_t = m_dot * ((h_7 - h_8)/1000) # kW
    t_power_vals.append(W_dot_t)

    eta_th = (W_dot_t - W_dot_pump_coolant - W_dot_pump_rankine) / Q_H_dot
    eta_vals.append(eta_th)

    W_net = W_dot_t - W_dot_pump_coolant - W_dot_pump_rankine
    net_power_vals.append(W_net)

fig, ax1 = plt.subplots()

# Efficiency axis
ax1.plot(P_vals, eta_vals, color= blue )
ax1.set_xlabel ( Turbine Inlet Pressure (Pa) )
ax1.set_ylabel ( Thermal Efficiency )
ax1.grid(True)

# Second axis for quality
ax2 = ax1.twinx()
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```
ax2.plot(P_vals, x_vals, color = red )
ax2.axhline(y=0.92, color = red , linestyle='--')
ax2.set_ylabel ( Turbine Exit Quality )
plt.show()

# Convert to numpy arrays (important for indexing)
eta_vals = np.array(eta_vals)
x_vals = np.array(x_vals)

# Find index of maximum efficiency
# Convert all to numpy arrays first
eta_vals = np.array(eta_vals)
x_vals = np.array(x_vals)
m_dot_vals = np.array(m_dot_vals)
m_dot_coolant_vals = np.array(m_dot_coolant_vals)
t_power_vals = np.array(t_power_vals)
rankine_pump_vals = np.array(rankine_pump_vals)
coolant_pump_vals = np.array(coolant_pump_vals)
net_power_vals = np.array(net_power_vals)

# Apply moisture constraint
valid = x_vals ≥ min_x

if not np.any(valid):
    raise ValueError ( No operating point satisfies the minimum quality ...
        constraint. )

# Filter valid region
eta_valid = eta_vals[valid]
P_valid = P_vals[valid]

idx_local = np.argmax(eta_valid)
idx_max = np.where(valid)[0][idx_local]

# Extract values
P_opt = P_vals[idx_max]
eta_max = eta_vals[idx_max]
x_opt = x_vals[idx_max]
m_dot_opt = m_dot_vals[idx_max]
m_dot_coolant_opt = m_dot_coolant_vals[idx_max]
t_power_opt = t_power_vals[idx_max]
net_power_opt = net_power_vals[idx_max]
W_dot_rankine_pump = rankine_pump_vals[idx_max]
W_dot_coolant_pump = coolant_pump_vals[idx_max]

print(f P_opt = {P_opt:.2f} Pa )
print(f eta_max = {eta_max:.4f} )
print(f Exit Quality at P_opt = {x_opt:.4f} )
print(f Rankine Mass Flow Rate = {m_dot_opt:.4f} kg/s )
print(f Coolant Mass Flow Rate = {m_dot_coolant_opt:.4f} kg/s )
print(f Turbine Power Output = {t_power_opt:.4f} kW )
print(f Net Power = {net_power_opt:.4f} kW )
print(f Power Input Rankine Pump = {W_dot_rankine_pump:.4f} kW )
print(f Power Input Coolant Pump = {W_dot_coolant_pump:.4f} kW )
```